

Engineering Curves: Construction of Cycloid

Rolling Circle Method with Tangent & Normal

Gajavada Sanjeevkumar

Problem Statement

Question

Draw a **cycloid** generated by a circle of diameter $D = 50$ mm rolling along a straight base line for **one complete revolution** without slipping.

Construct a **tangent** and a **normal** to the cycloid at any point P located on the curve.

Step 1: Draw Generating Circle & Directing Line (Base)



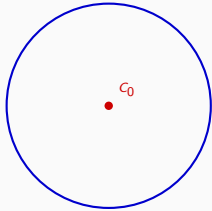
C_0

$C_0 \rightarrow$ Initial Center Point

$AA' \rightarrow$ Directing Line

Axis $\rightarrow$ Center Line

Step 1: Draw Generating Circle & Directing Line

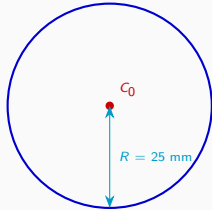


$C_0 \rightarrow$ Initial Center Point

$AA' \rightarrow$ Directing Line

Axis $\rightarrow$ Center Line

Step 1: Draw Generating Circle & Directing Line



C_0 → Initial Center Point
 AA' → Directing Line
Axis → Center Line

Step 1: Draw Generating Circle & Directing Line

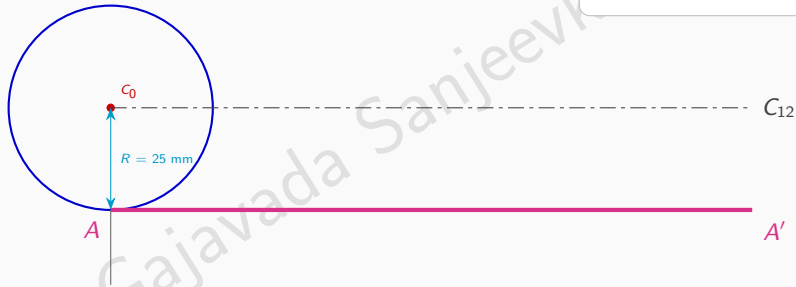


C_0 → Initial Center Point
 AA' → Directing Line
Axis → Center Line

Step 1: Draw Generating Circle & Directing Line



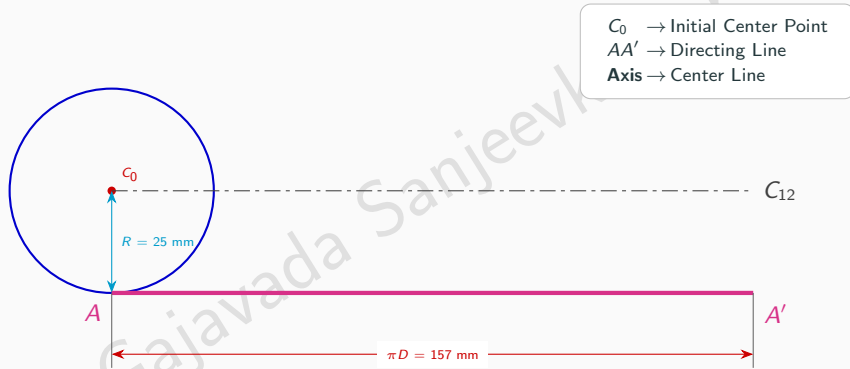
Step 1: Draw Generating Circle & Directing Line



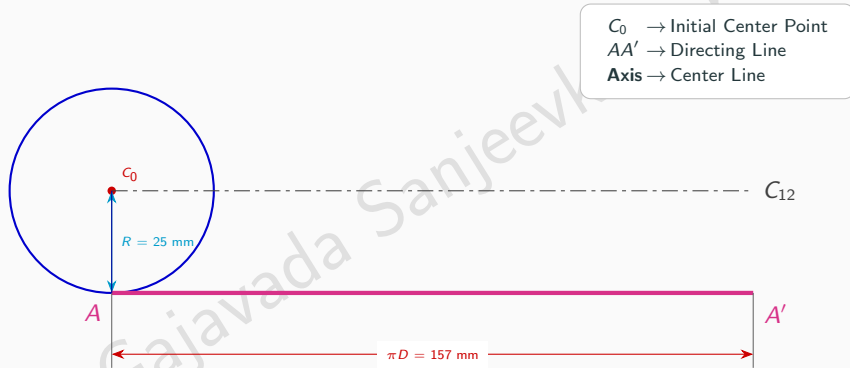
Step 1: Draw Generating Circle & Directing Line



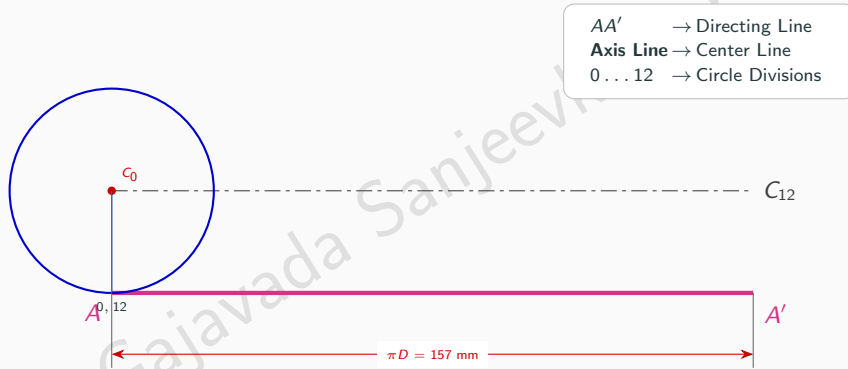
Step 1: Draw Generating Circle & Directing Line



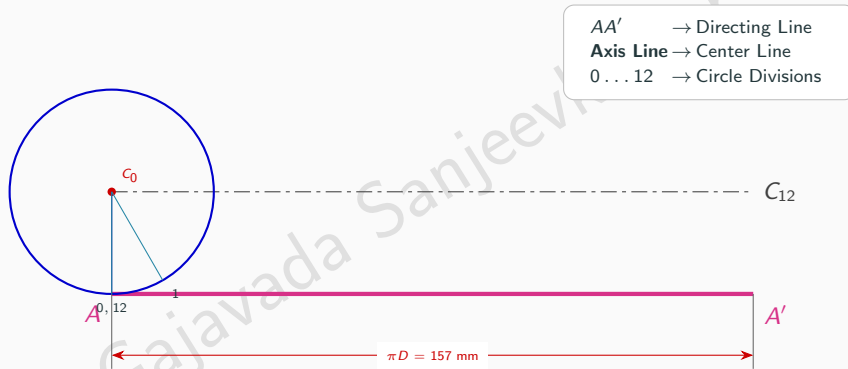
Step 1: Draw Generating Circle & Directing Line



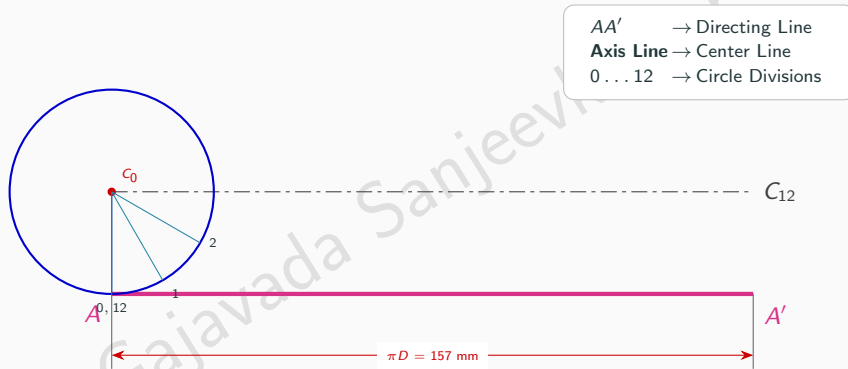
Step 2: Divide Generating Circle into 12 Equal Parts



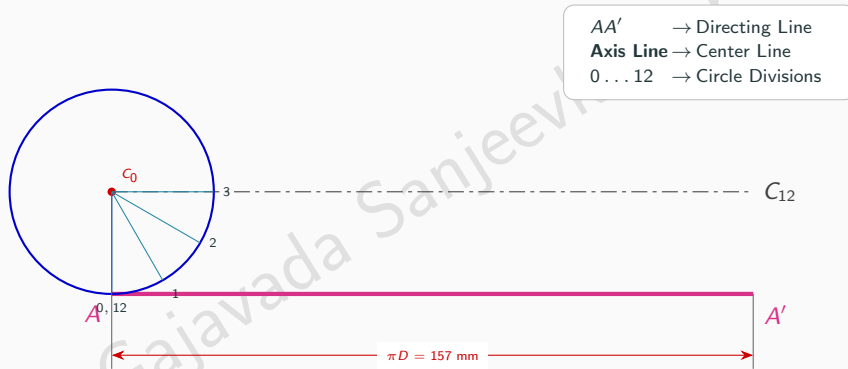
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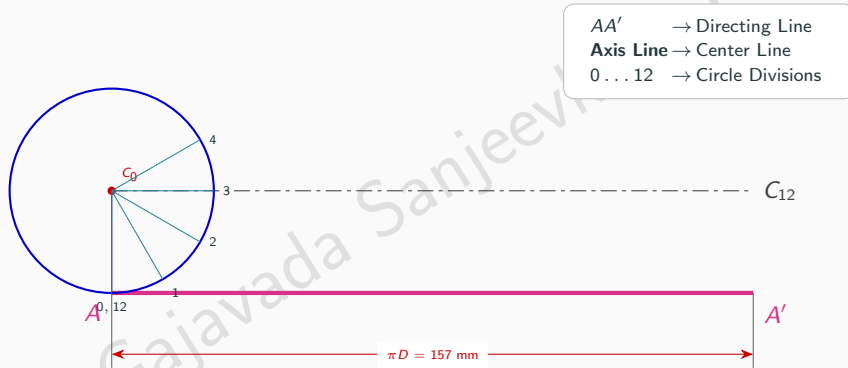
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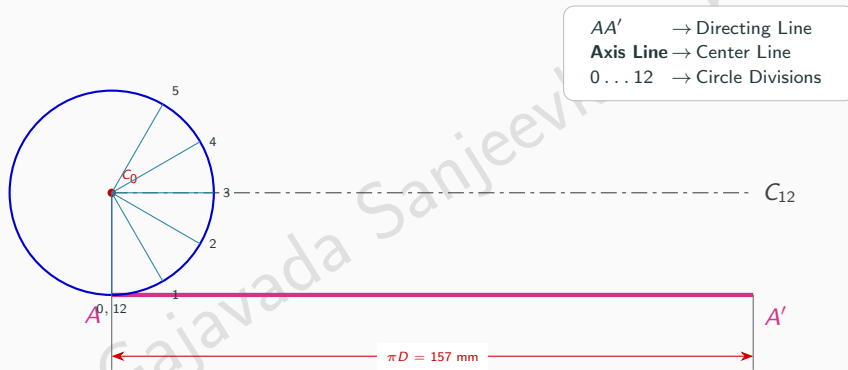
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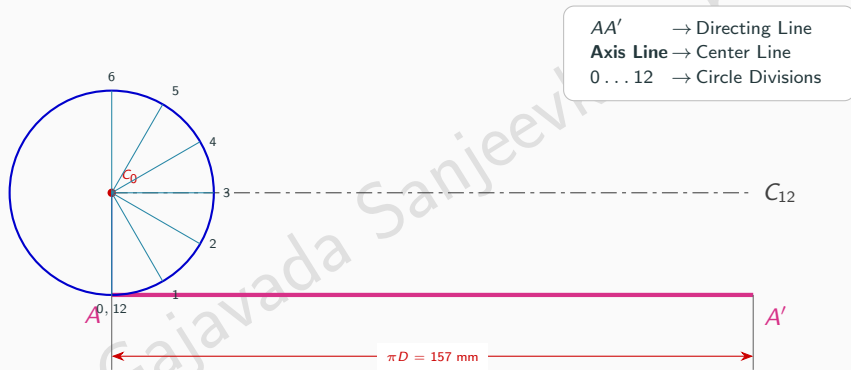
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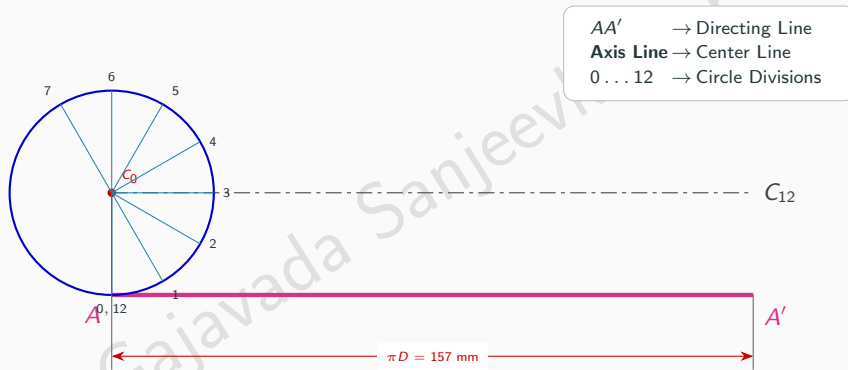
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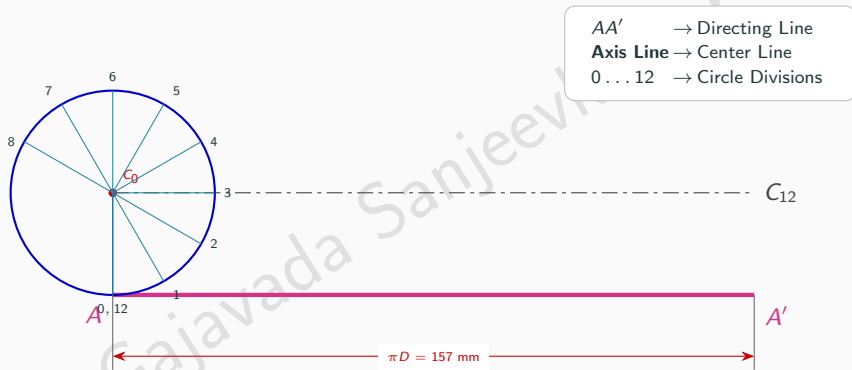
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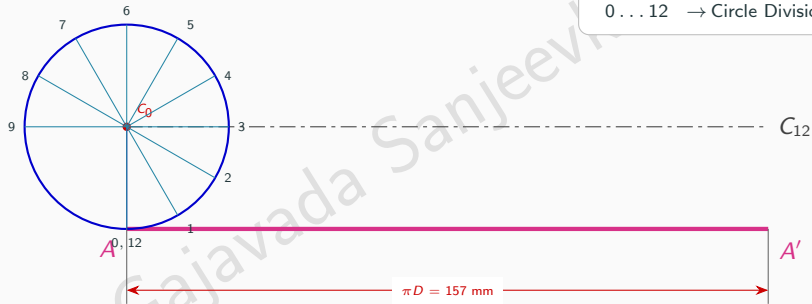
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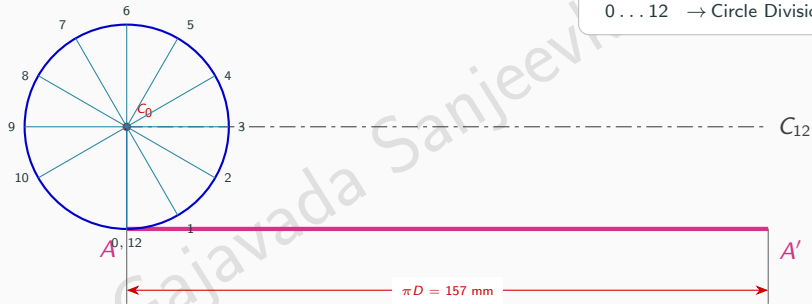
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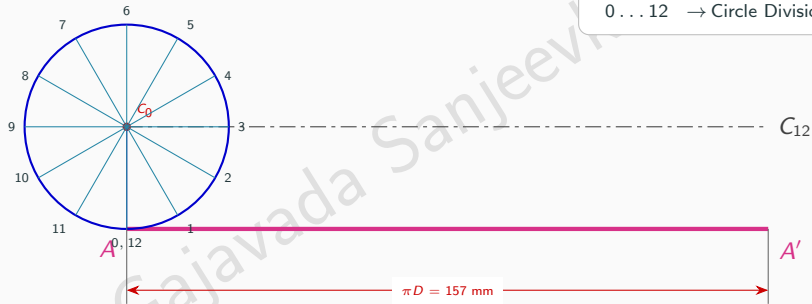
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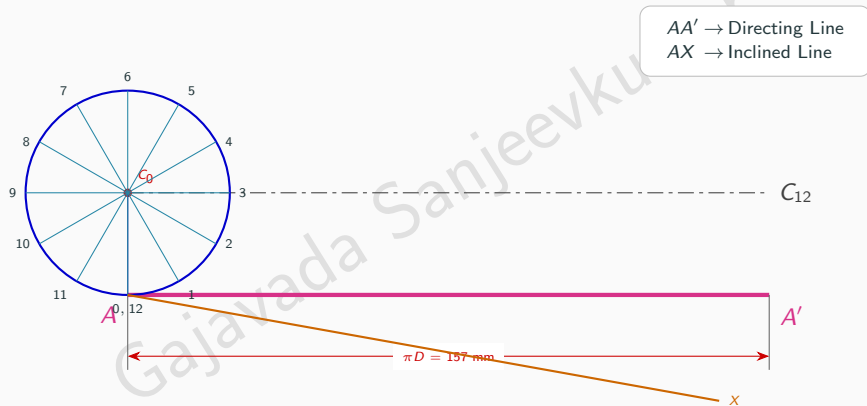
Step 2: Divide Generating Circle into 12 Equal Parts



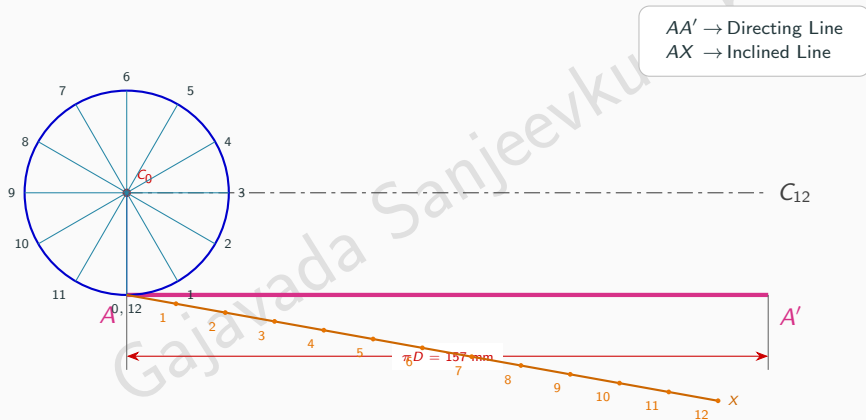
Step 2: Divide Generating Circle into 12 Equal Parts



Step 3: Divide Directing Line - Setup Inclined Line AX

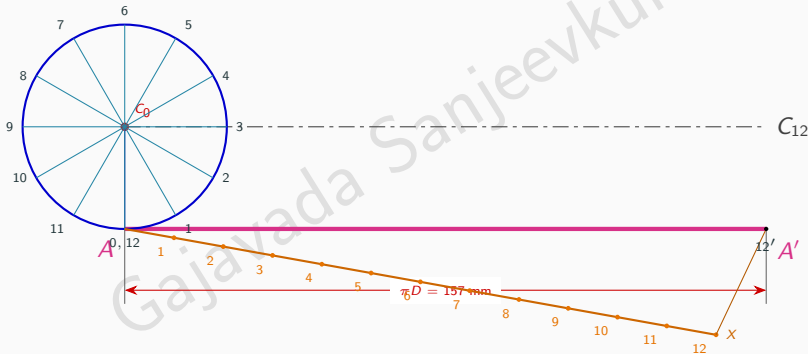


Step 3: Divide Directing Line - Add Numbers on Line AX



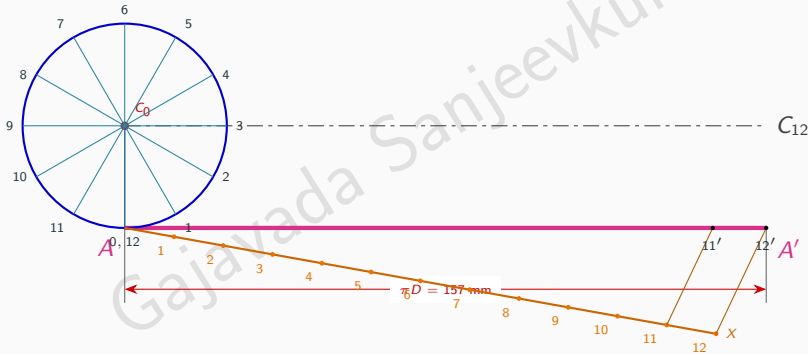
Step 3: Divide Directing Line - Transfer Point 12'

12' → Transfer Point 12



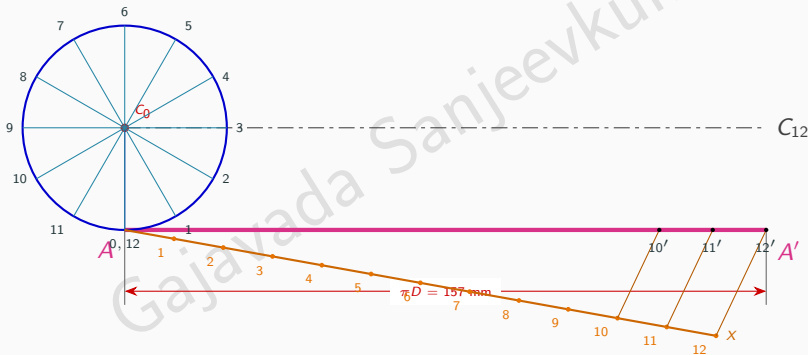
Step 3: Divide Directing Line - Transfer Point 11'

11' → Transfer Point 11

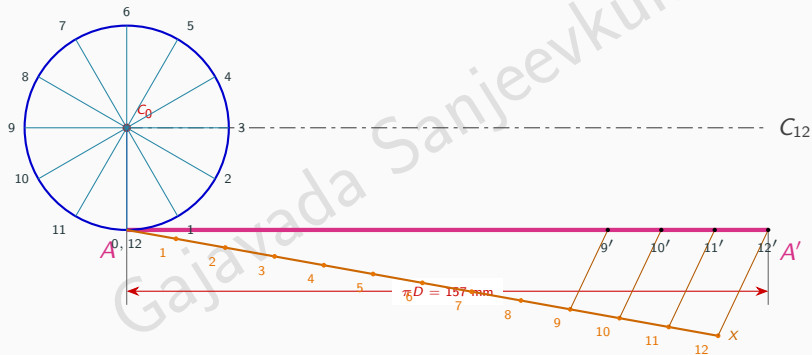


Step 3: Divide Directing Line - Transfer Point 10'

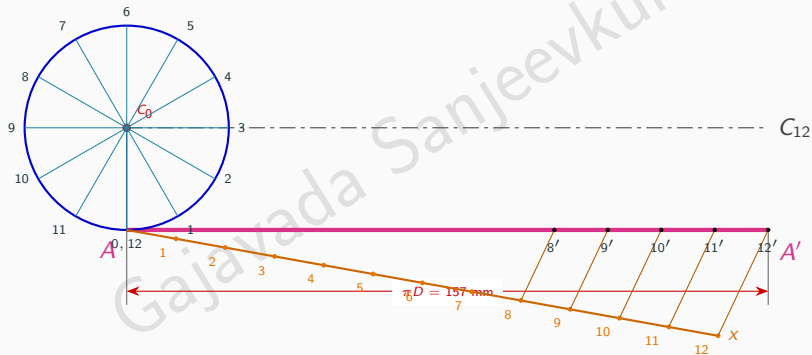
10' → Transfer Point 10



Step 3: Divide Directing Line - Transfer Point 9'

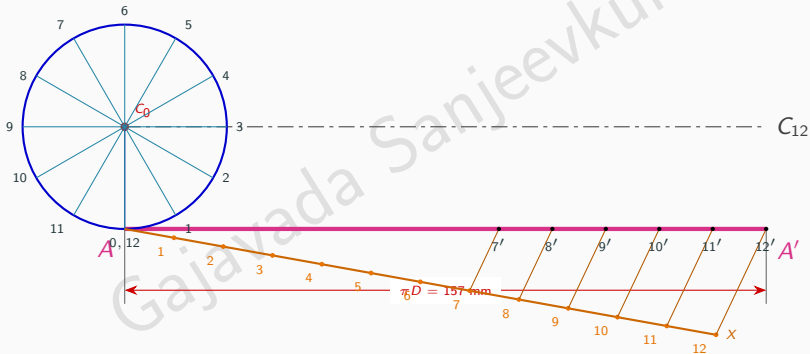


Step 3: Divide Directing Line - Transfer Point 8'



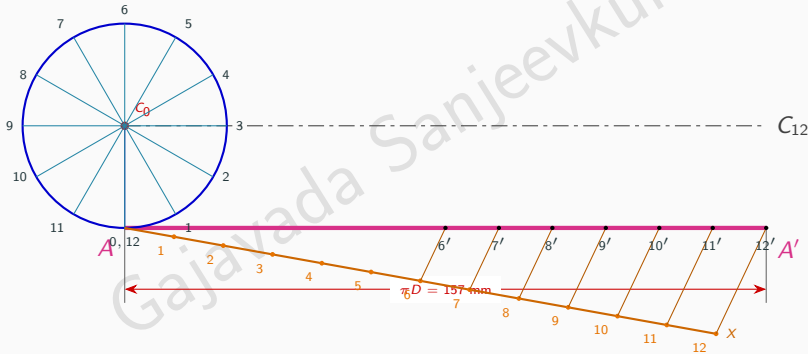
Step 3: Divide Directing Line - Transfer Point 7'

7' → Transfer Point 7



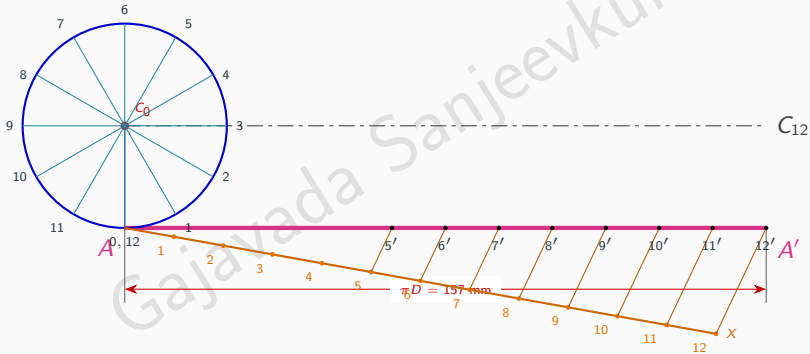
Step 3: Divide Directing Line - Transfer Point 6'

6' → Transfer Point 6



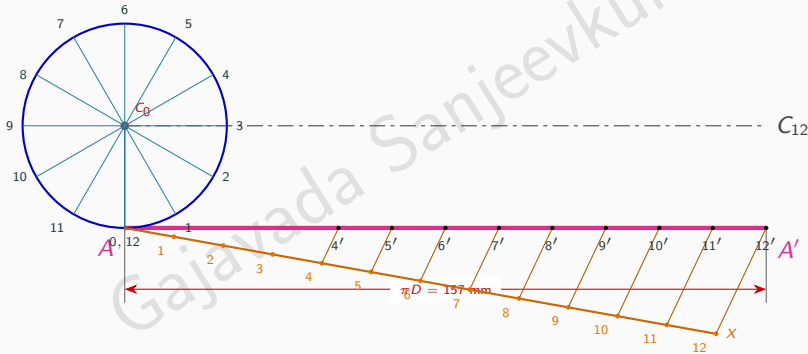
Step 3: Divide Directing Line - Transfer Point 5'

5' → Transfer Point 5



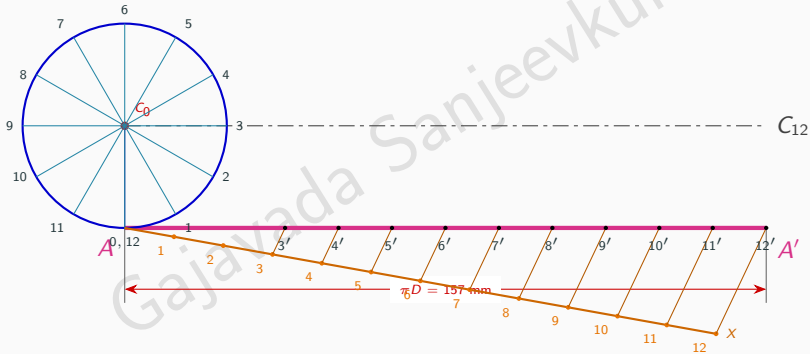
Step 3: Divide Directing Line - Transfer Point 4'

4' → Transfer Point 4



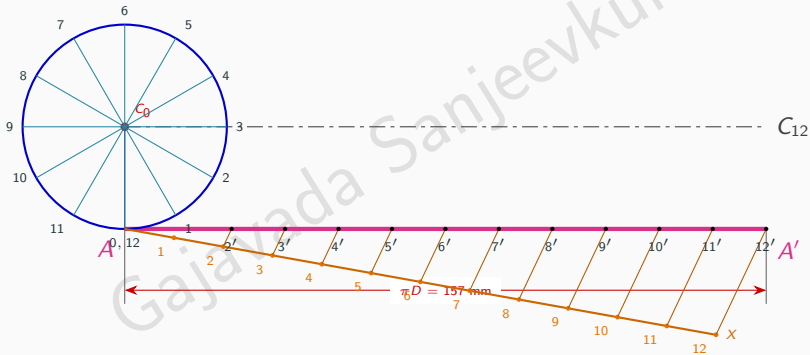
Step 3: Divide Directing Line - Transfer Point 3'

3' → Transfer Point 3



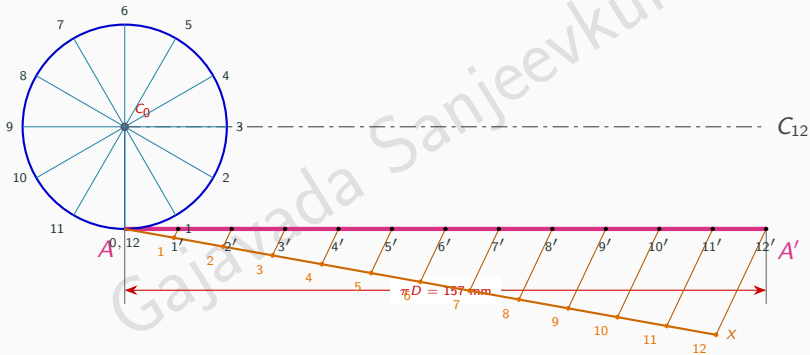
Step 3: Divide Directing Line - Transfer Point 2'

2' → Transfer Point 2

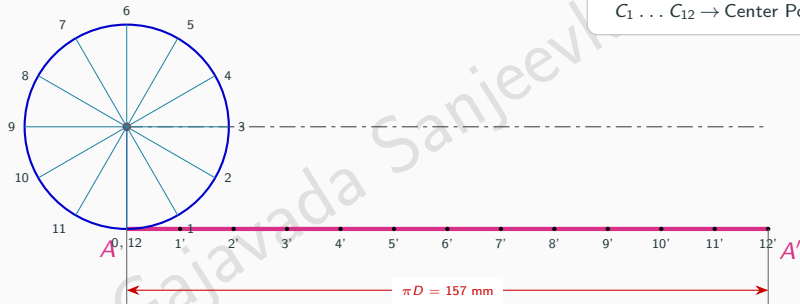


Step 3: Divide Directing Line - Transfer Point 1'

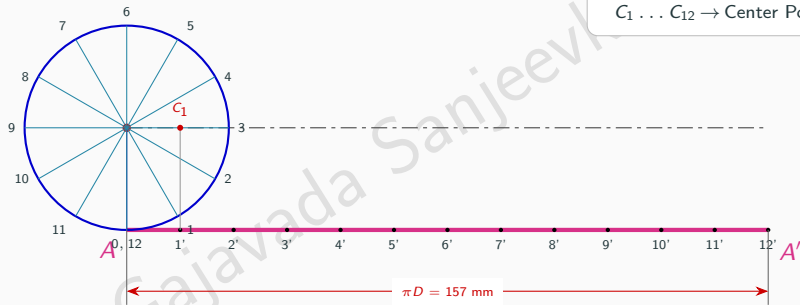
1' → Transfer Point 1



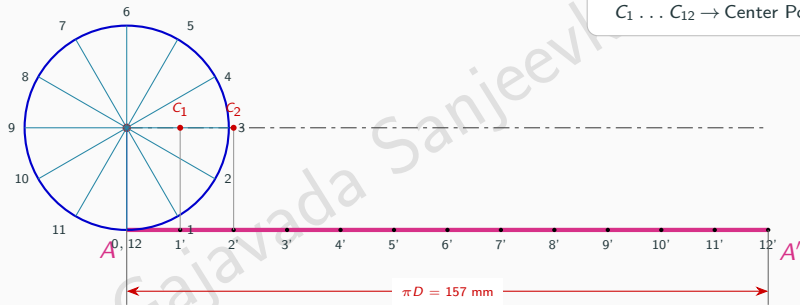
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$ (Base)



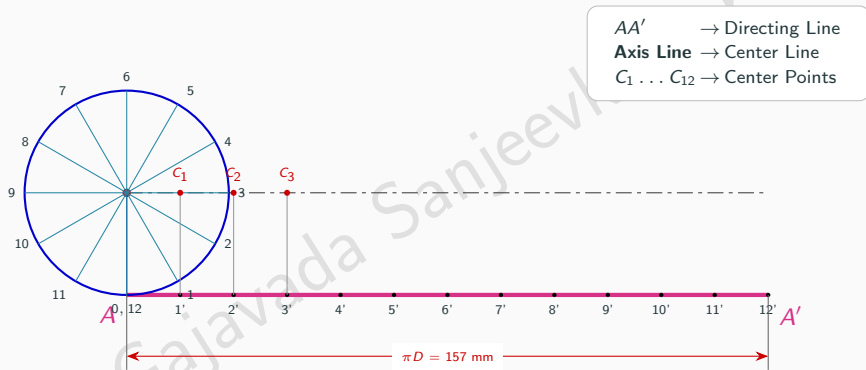
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



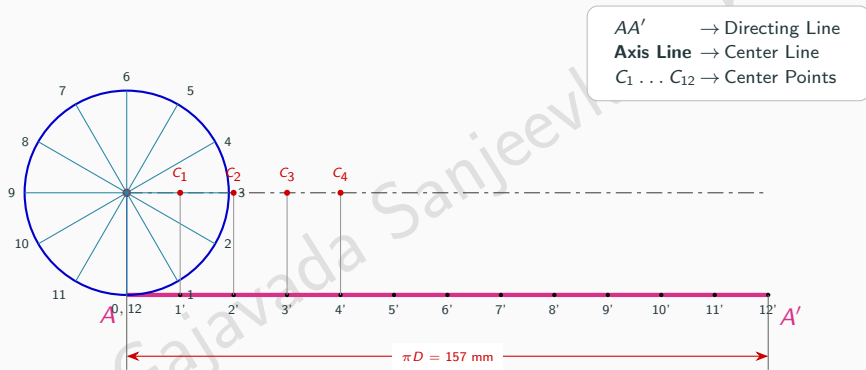
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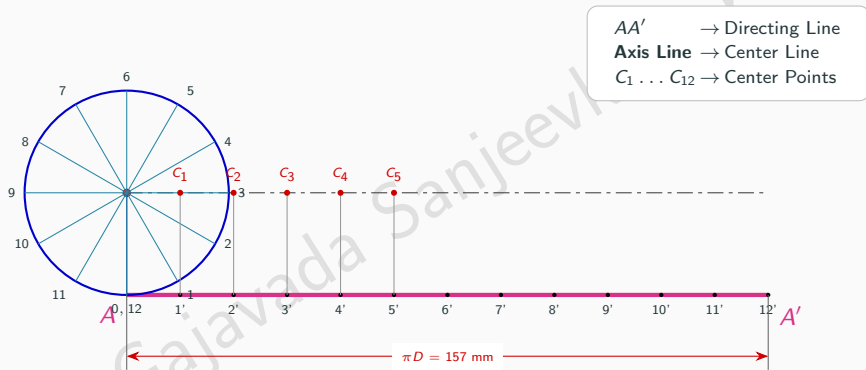
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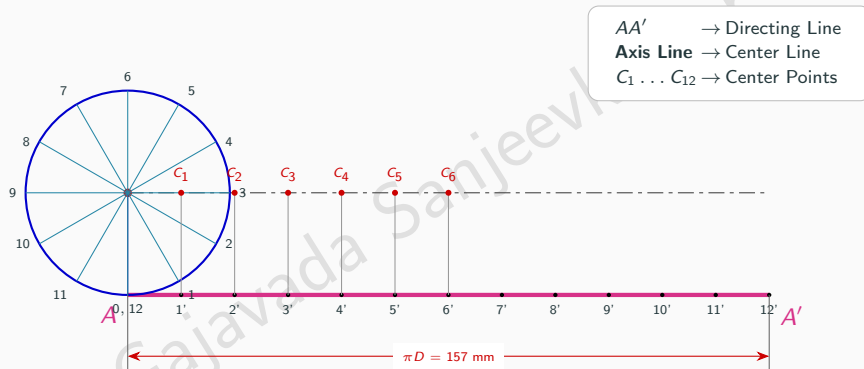
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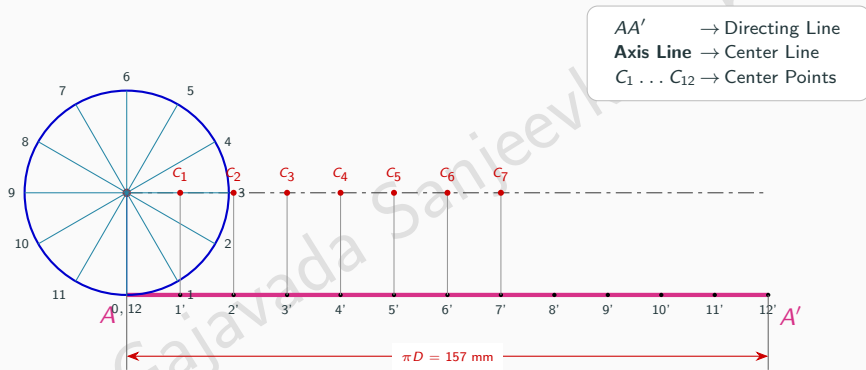
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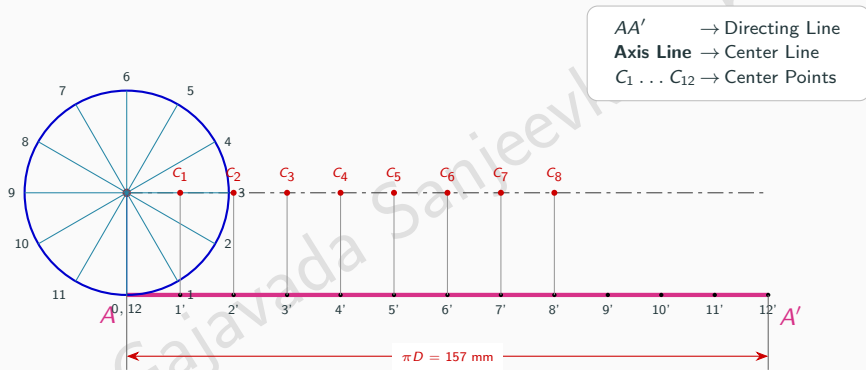
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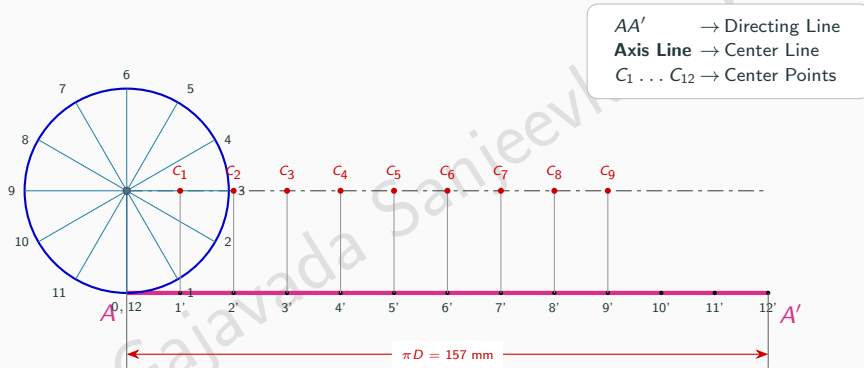
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



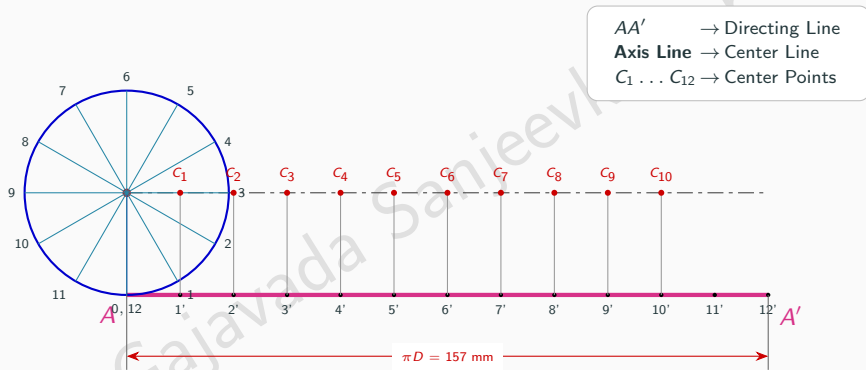
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



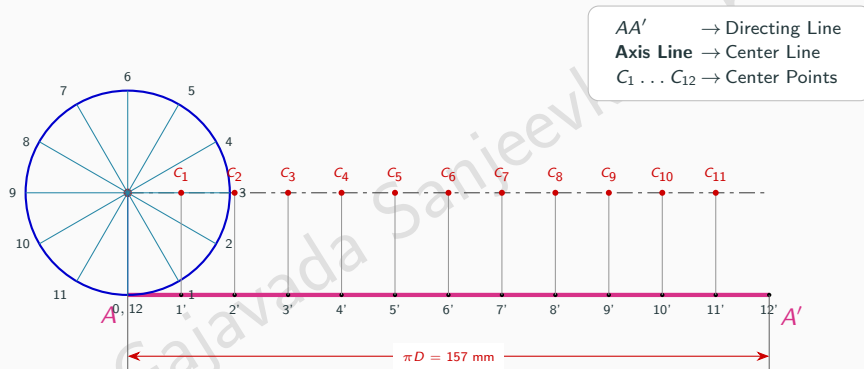
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



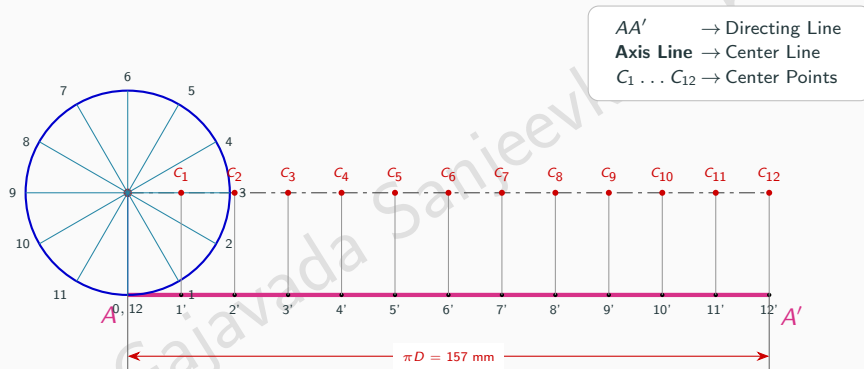
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



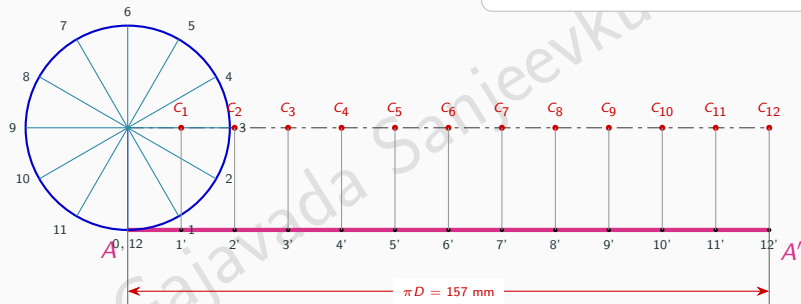
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



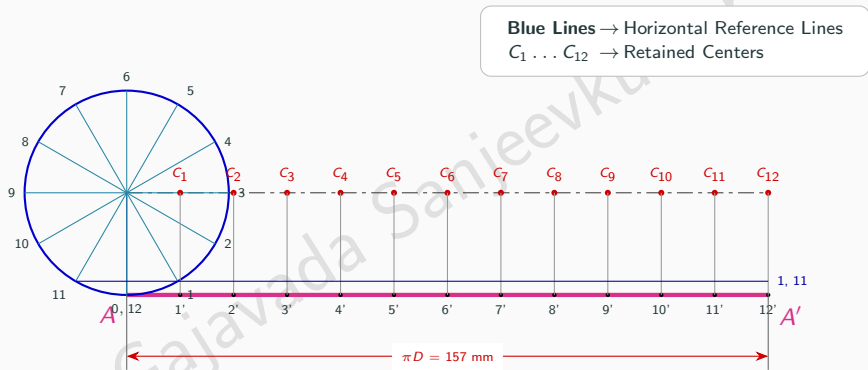
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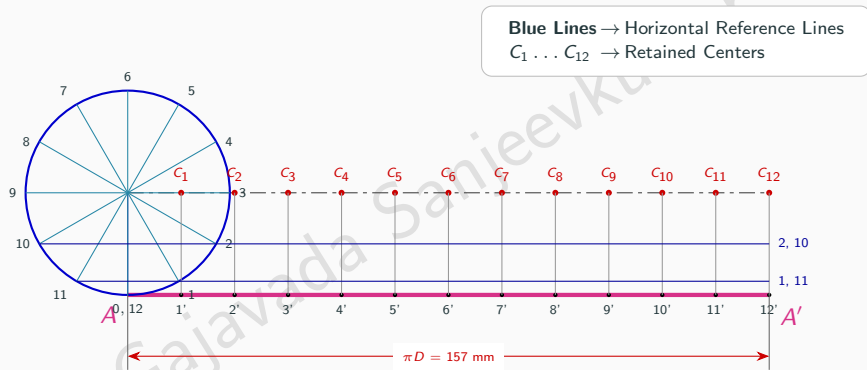
Step 5: Draw Horizontal Reference Lines (Base)



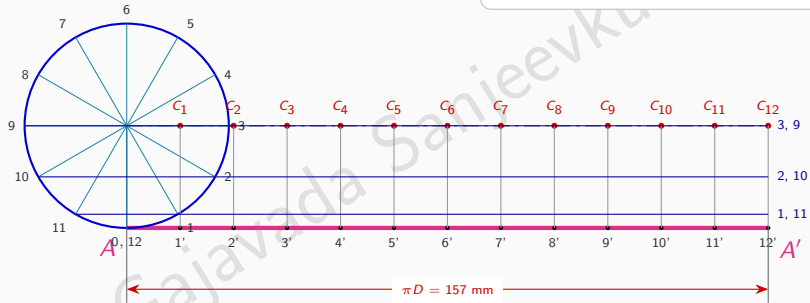
Step 5: Draw Horizontal Reference Lines



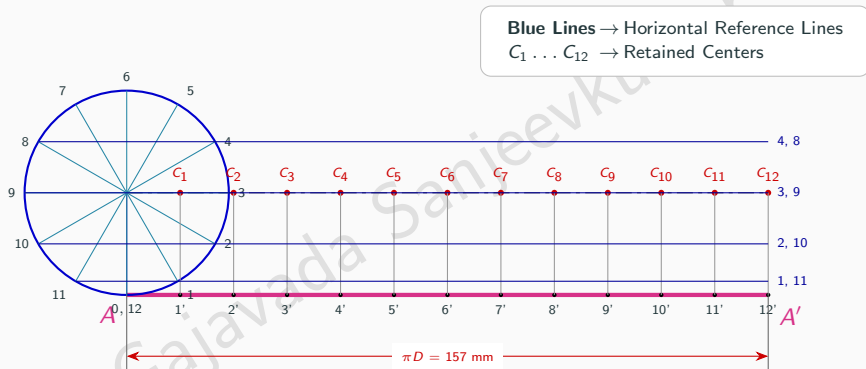
Step 5: Draw Horizontal Reference Lines



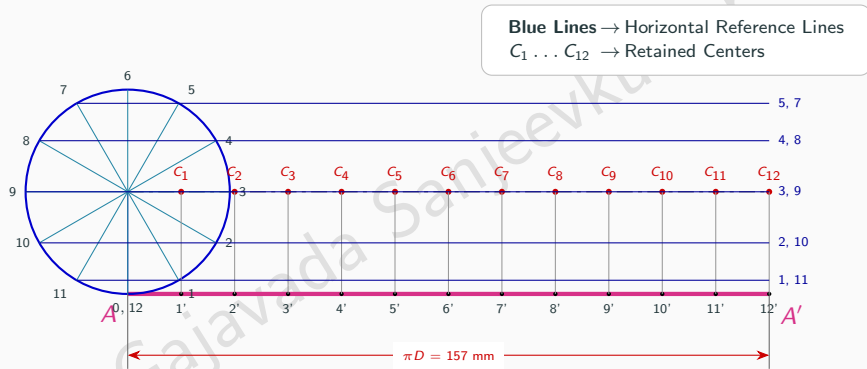
Step 5: Draw Horizontal Reference Lines



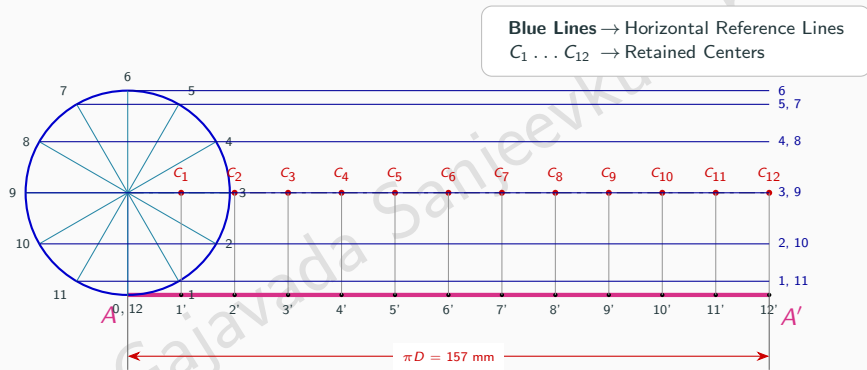
Step 5: Draw Horizontal Reference Lines



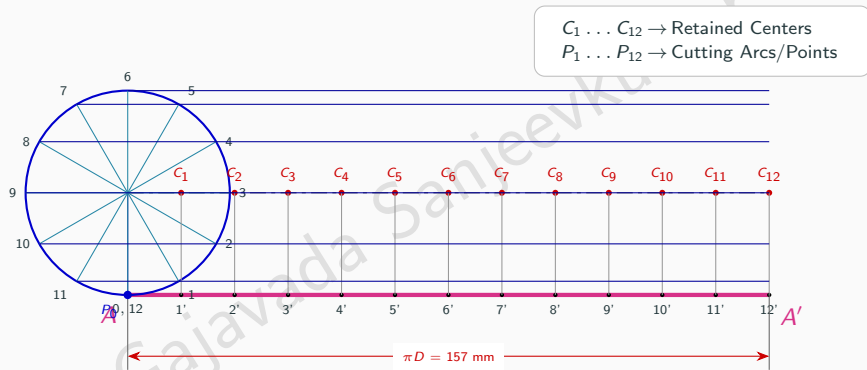
Step 5: Draw Horizontal Reference Lines



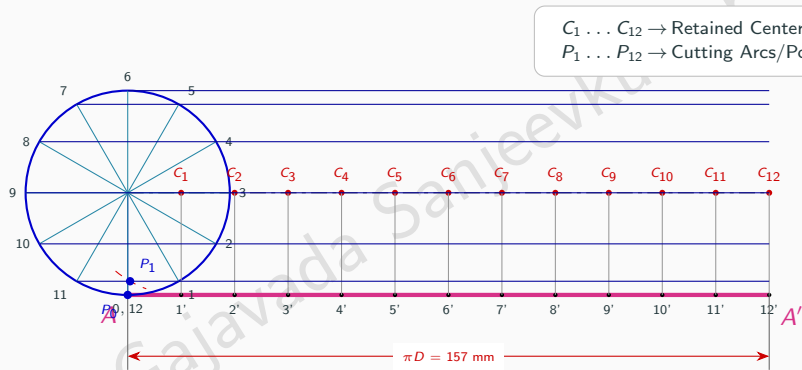
Step 5: Draw Horizontal Reference Lines



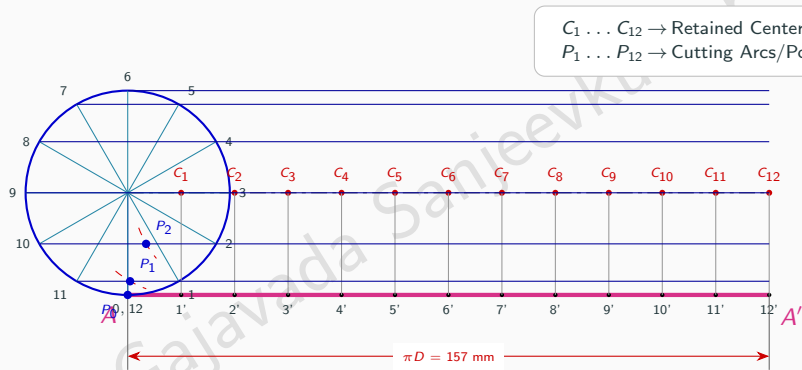
Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate P_0



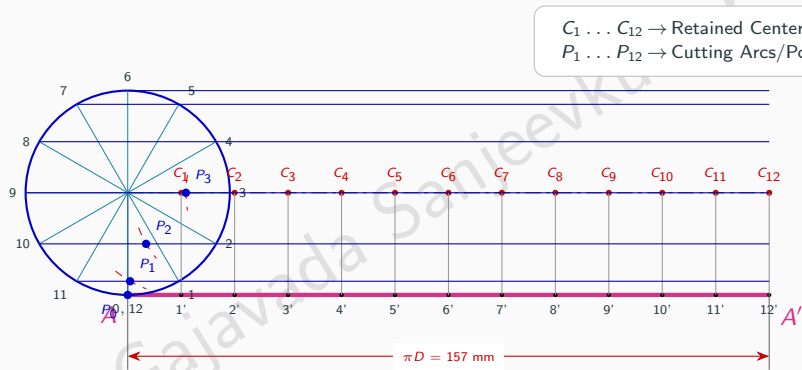
Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate P_1



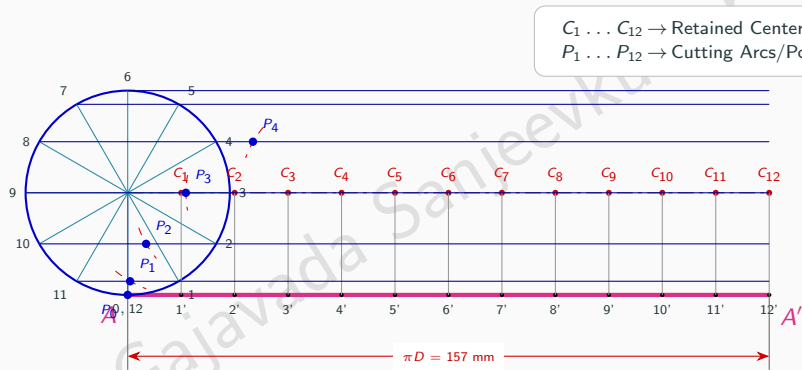
Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate P_2



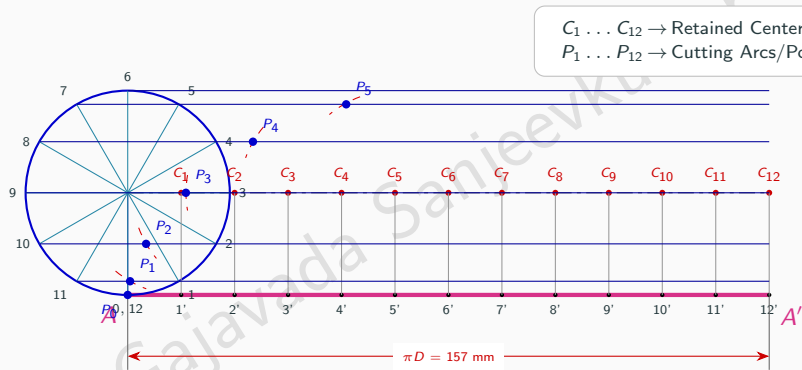
Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate P_3



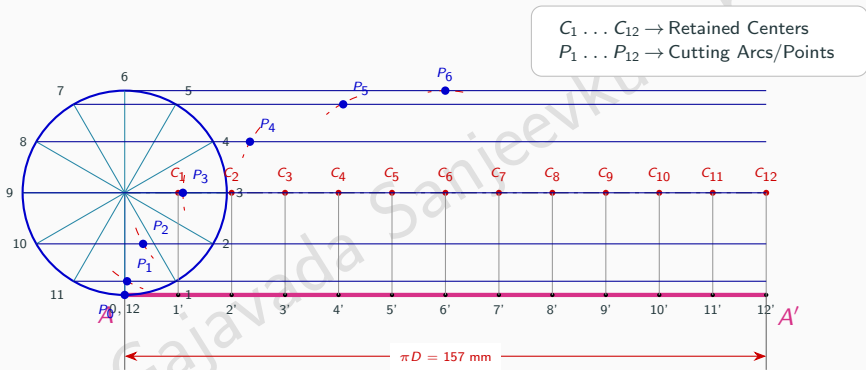
Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate P_4



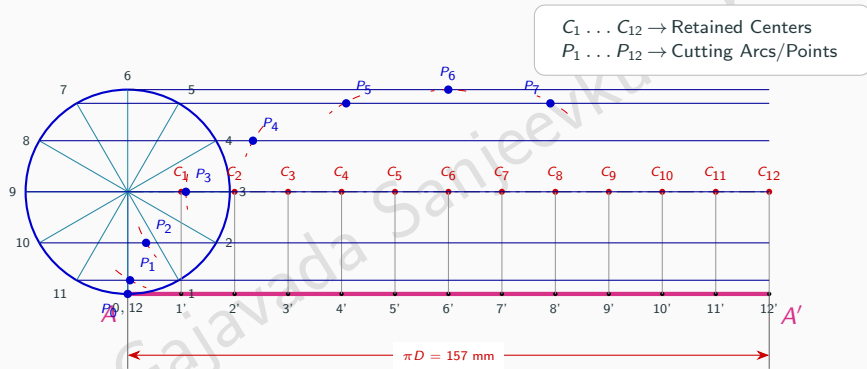
Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate P_5



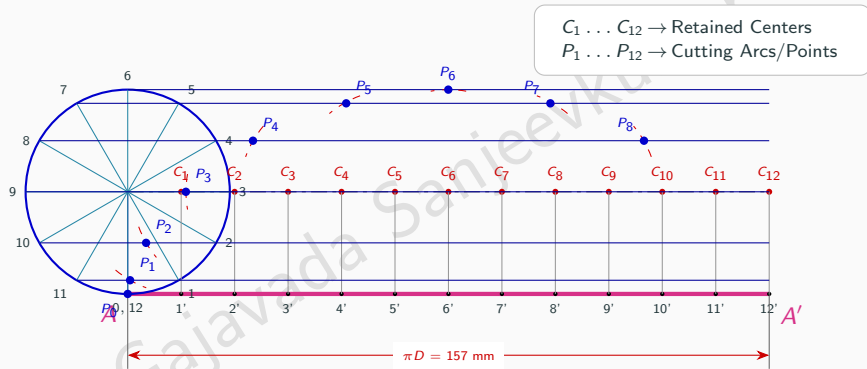
Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate P_6



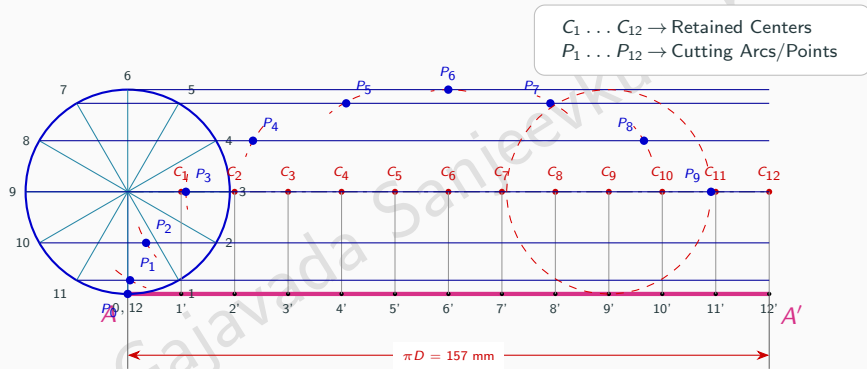
Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate P_7



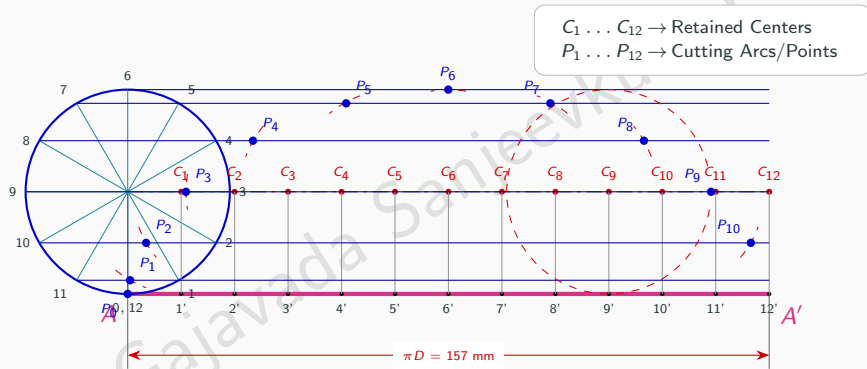
Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate P_8



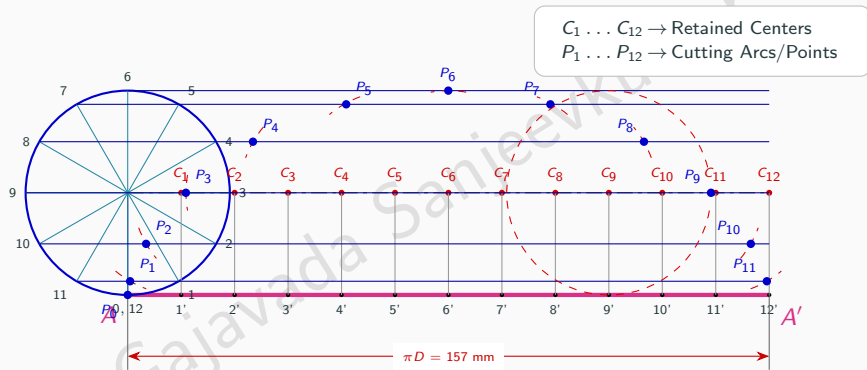
Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate P_9



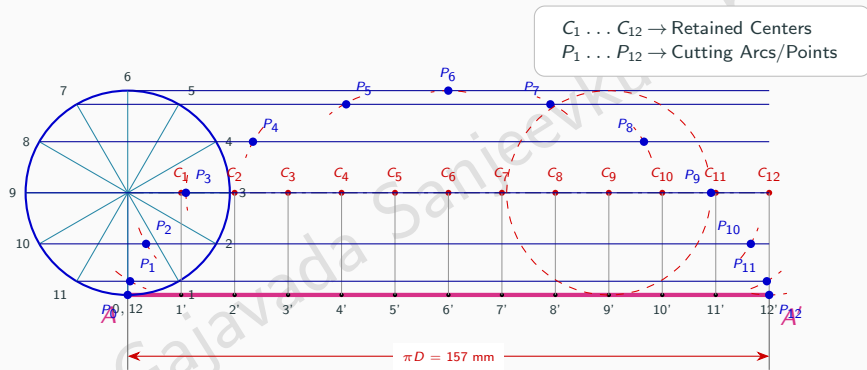
Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate P_{10}



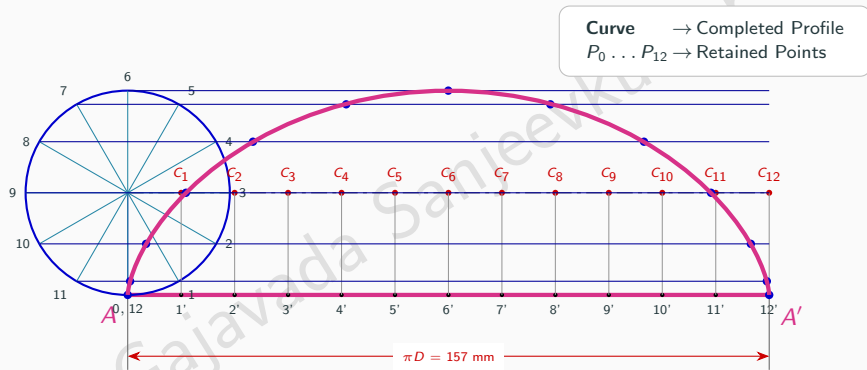
Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate P_{11}



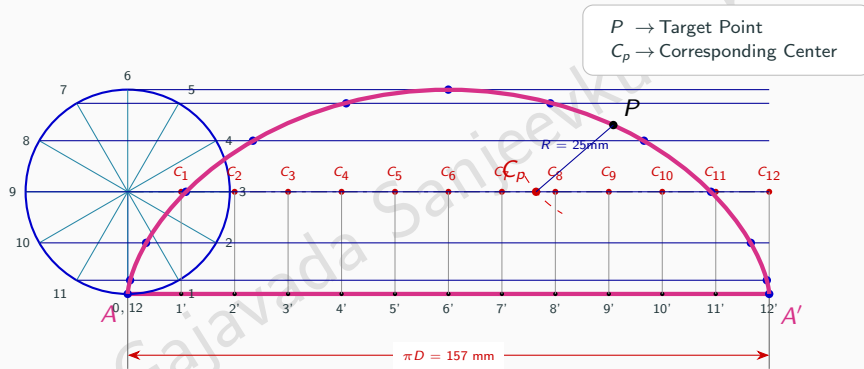
Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate P_{12}



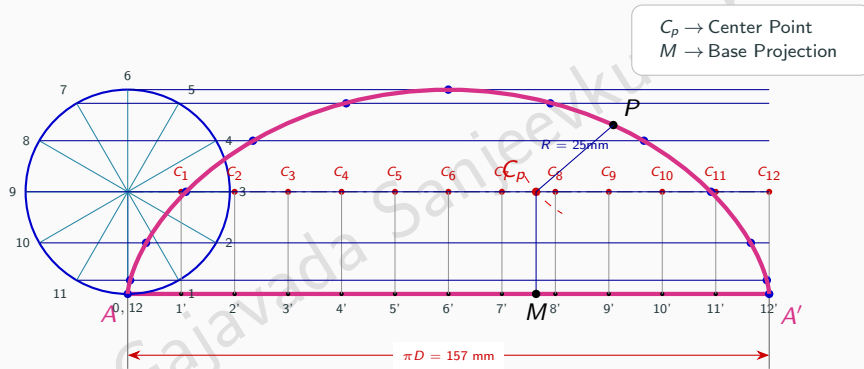
Step 7: Smoothly Join Points to Form Cycloid Curve



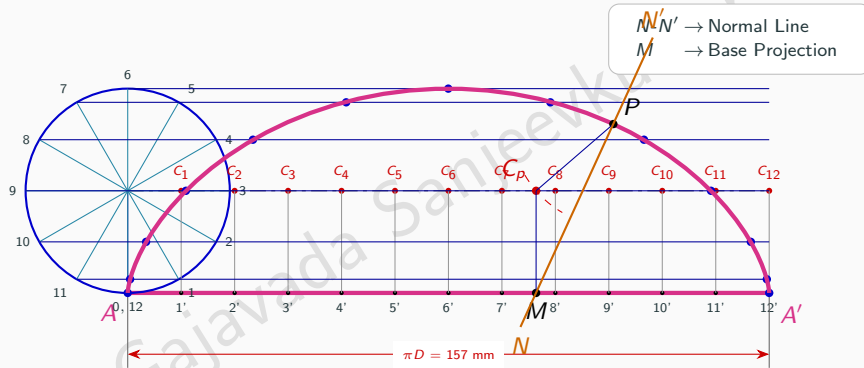
Step 8: Mark Point P (Right Side) & Locate Center C_p



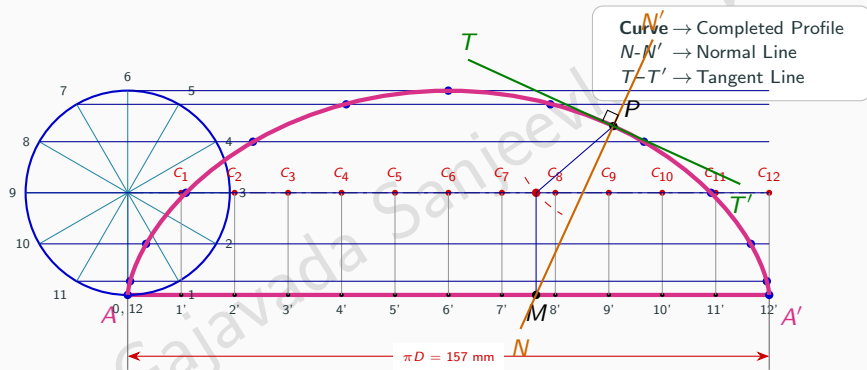
Step 9a: Project C_p onto Directing Line and Mark M



Step 9b: Construct Normal Line $N-N'$ through M and P



Step 10: Construct Tangent Line $T-T'$ & Final Result



Thank You

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Department of Mechanical Engineering

IIT (ISM) Dhanbad

For academic queries, teaching material, or further information:

`skgajawada.github.io/#/contact`

Engineering Graphics — Cycloid Construction